

■ Global Jersey Store

Technical Project Report

E-Commerce Application for International Football Jerseys

Nossaiba El Asri
July 2026

-
- Live: <https://global-jersey-store.vercel.app>
 - GitHub: <https://github.com/nossaiba-elas/global-jersey-store>

1. Introduction

Global Jersey Store is a modern, full-featured e-commerce web application specialised in the sale of international football jerseys. Developed as a professional portfolio project, it showcases advanced skills in modern web development, covering the full spectrum from UI/UX design to authentication, state management, and CI/CD deployment.

Primary Objectives

- Demonstrate end-to-end full-stack web development competency.
- Implement a premium, production-grade user experience.
- Build a complete shopping flow: browse → cart → checkout → order tracking.
- Apply modern DevOps practices (CI/CD, automated testing, cloud deployment).

Technology Stack

Layer	Technology
Framework	Next.js 15 (App Router)
Language	TypeScript (strict mode)
Styling	Tailwind CSS + shadcn/ui + @base-ui/react
State Management	Zustand (with localStorage persistence)
Authentication	Firebase Google OAuth
Deployment	Vercel (auto-deploy on push)
CI/CD	GitHub Actions (lint + build)
Version Control	GitHub

2. Technical Architecture

2.1 Frontend — Next.js 15 App Router

The application leverages Next.js 15 with the App Router paradigm, combining Server Components for fast initial loads and Client Components for interactive UI elements. The file-based routing system organises pages under the `app/` directory, covering Home, Shop, Product Detail, Cart, Checkout, Profile, Wishlist, Login, and Admin.

2.2 State Management — Zustand

Four dedicated Zustand stores handle application state with full `localStorage` persistence, ensuring data survives page refreshes:

- **cart-store** — Shopping cart items and quantities.
- **wishlist-store** — User's saved/favourite products.
- **orders-store** — Order history with GJS-YYYY-XXXXXX reference numbers.
- **products-store** — Product catalogue shared between shop and admin panel.

2.3 Authentication — Firebase

Google OAuth is handled via Firebase Authentication. The auth state is managed globally and gates access to the user Profile and Admin Dashboard pages. The admin panel is restricted to a specific authorised email address, preventing unauthorised access.

2.4 Component Architecture

The project follows a feature-based folder structure. Shared UI primitives (Button, Card, Badge, Dialog, Sheet, etc.) live in *components/ui/*. Feature-specific logic is co-located under *features/* (shop, cart, checkout, admin, product, home). The *@base-ui/react* library is used for accessible composable primitives, using the *render* prop pattern for composition.

3. Key Features

3.1 Product Catalogue

- 20+ international football jerseys (home, away, third kits) with real team images.
- Advanced filtering: country, brand, size, price range.
- Sorting: newest, price ascending/descending, popularity.
- Paginated results (8 per page) with skeleton loading states.
- Product badges: ■ Trending, Featured, Out of Stock.

3.2 Shopping Experience

- Persistent cart with quantity management and real-time totals.
- Wishlist to save favourite jerseys across sessions.
- Multi-step checkout with shipping address form.
- Instant search overlay accessible from the header.

3.3 Order Management

- Unique order references in the format GJS-2026-XXXXXX.
- Full order history on the user profile page.
- Status tracking: Confirmed → Processing → Shipped → Delivered.
- Itemised order breakdown with pricing, tax, and delivery details.

3.4 Admin Dashboard

- Restricted access — only the registered admin email can view this page.
- Overview statistics: total sales, orders, products, out-of-stock count.
- Product management: inline stock editing, price updates, product deletion.
- Order management: update order status (Confirmed → Delivered).
- Analytics tab: order status breakdown and top-selling products.

3.5 UI/UX Design

- Premium dark-themed aesthetic inspired by professional sports retail (World Soccer Shop).
- Mobile-first responsive design across all screen sizes.
- Light/dark mode toggle persisted in localStorage.
- Custom shield-style logo and brand identity.

4. CI/CD Pipeline

Every push to the **main** branch triggers an automated GitHub Actions workflow that runs two jobs in parallel:

- **Lint** — ESLint checks for code quality and style consistency.
- **Build** — Full Next.js production build to verify zero compilation errors.

On successful merge to main, Vercel automatically deploys the latest version to production at <https://global-jersey-store.vercel.app>, ensuring zero-downtime continuous deployment.

5. Technical Challenges & Solutions

Challenge	Solution
@base-ui/react vs Radix UI API	Systematically replaced all 'asChild' props with the required 'render' prop across 9+ components
Firebase auth domain on Vercel	Registered production domain in Firebase Console authorised domains list
Shared product state (Admin ↔ Shop)	Implemented shared Zustand products-store with localStorage persistence
Image handling for custom products	Removed Add Product feature; admin manages existing catalogue only — clean for porting
TypeScript strict mode	Full type coverage across stores, components, and API boundaries

6. Skills Demonstrated

- **React & Next.js 15** — App Router, Server/Client Components, dynamic routing.
- **TypeScript** — Strict typing across all layers (stores, props, API boundaries).
- **State Architecture** — Zustand stores with persistence middleware.
- **OAuth Integration** — Firebase Google Sign-In with protected routes.
- **CI/CD** — GitHub Actions + Vercel for automated testing and deployment.
- **UI/UX** — Tailwind CSS, responsive design, dark/light mode, accessibility.
- **Git & Collaboration** — Feature branches, commit conventions, pull requests.

7. Conclusion

Global Jersey Store demonstrates full mastery of the modern React/Next.js ecosystem, from architecture and component design to authentication, state management, and cloud deployment. It reflects the ability to independently design, build, and ship a production-ready web application — the exact skill set sought by professional development teams.

- **Live Application:** <https://global-jersey-store.vercel.app>
- **Source Code:** <https://github.com/nossaiba-elas/global-jersey-store>